

Rigid Analytic Geometry Week 1 Solution

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Problem 1

By multiplicative property of the absolute value, we have K° is a valuation ring of K . Thus $\text{Spec } K^\circ = \{K^{\circ\circ}, \{0\}\}$.

Problem 2

The second assertion is clear as for any $\alpha \in \mathbb{Z}_{\geq 0}^m$, $1_\alpha = 0$ unless $\alpha = 0$. For the first assertion, from the definition of $\|\cdot\|_{\mathbb{T}_m}$, for each $f, g \in \mathbb{T}_m$ there are $\alpha, \beta \in \mathbb{Z}_{\geq 0}^m$ such that

$$|f_\alpha| = \|f\|_{\mathbb{T}_m}, |g_\beta| = \|g\|_{\mathbb{T}_m}. \quad (1)$$

Also,

$$\begin{aligned} \|fg\|_{\mathbb{T}_m} &= \max_{\gamma} \left\{ \left| \sum_{\alpha+\beta=\gamma} f_\alpha g_\beta \right| \mid \gamma \in \mathbb{Z}_{\geq 0}^m \right\}, \\ &\leq |f_\alpha| |g_\beta|, \\ &= \|f\|_{\mathbb{T}_m} \|g\|_{\mathbb{T}_m}. \end{aligned}$$

To show multiplicativeness, consider,

$$\alpha_0 = \min\{\alpha \mid |f_\alpha| = \|f\|_{\mathbb{T}_m}\}, \beta_0 = \min\{\beta \mid |g_\beta| = \|g\|_{\mathbb{T}_m}\}.$$

Then,

$$|(fg)_{\alpha_0+\beta_0}| \leq \|fg\|_{\mathbb{T}_m}.$$

For any (α, β) with $\alpha + \beta = \alpha_0 + \beta_0$ we have,

$$|f_\alpha g_\beta| < |f_{\alpha_0} g_{\beta_0}|.$$

As $|\cdot|$ is non-archimedian, we have,

$$\left| \sum_{\substack{\alpha+\beta=\alpha_0+\beta_0 \\ (\alpha,\beta) \neq (\alpha_0,\beta_0)}} f_\alpha g_\beta \right| < |f_{\alpha_0} g_{\beta_0}|.$$

This can be seen by induction. Using it is non-archimedian, we have,

$$\begin{aligned} |f_{\alpha_0}g_{\beta_0}| &= \left| (fg)_{\alpha_0+\beta_0} - \sum_{\substack{\alpha+\beta=\alpha_0+\beta_0 \\ (\alpha,\beta) \neq (\alpha_0,\beta_0)}} f_{\alpha}g_{\beta} \right|, \\ &\leq \max \left\{ |(fg)_{\alpha_0+\beta_0}|, \left| \sum_{\substack{\alpha+\beta=\alpha_0+\beta_0 \\ (\alpha,\beta) \neq (\alpha_0,\beta_0)}} f_{\alpha}g_{\beta} \right| \right\} \end{aligned}$$

But

$$\left| \sum_{\substack{\alpha+\beta=\alpha_0+\beta_0 \\ (\alpha,\beta) \neq (\alpha_0,\beta_0)}} f_{\alpha}g_{\beta} \right| < |f_{\alpha_0}g_{\beta_0}|.$$

Thus we conclude

$$|(fg)_{\alpha_0+\beta_0}| = |f_{\alpha_0}g_{\beta_0}|,$$

showing this is multiplicative.

Problem 3

Clearly 0, 1 are idempotents thus $0, 1 \in A^\circ$. From the first property of the norm,

$$\|ab|A\| \leq \|a|A\| \|b|A\|,$$

$\{(ab)^n \mid n \in \mathbb{N}\}$ is bounded by the products of upperbounds of $\{a^n \mid n \in \mathbb{N}\}, \{b^n \mid n \in \mathbb{N}\}$, thus $ab \in A^\circ$. Similarly for $a + b$, as

$$\|(a+b)^n|A\| \leq \|a+b|A\|^n \leq \max\{\|a|A\|, \|b|A\|\}^n.$$

Thus we can take the maximum of the upperbounds of $\{a^n \mid n \in \mathbb{N}\}, \{b^n \mid n \in \mathbb{N}\}$.

If $a \in A^\circ$ and $x \in A^{\circ\circ}$ then as $x^n \xrightarrow{n \rightarrow \infty} 0$ and there is $M > 0$ such that $\|a|A\|^n \leq M$, we have,

$$\|(ax)^n|A\| \xrightarrow{n \rightarrow \infty} 0.$$

Since A is a complete metric space with respect to the induced metric from $\|\cdot|A\|$, 0 is the unique element of norm 0, thus $ax \in A^{\circ\circ}$. If $a^k \in A^{\circ\circ}$, then again by that A is a complete metric space we have $a \in A^{\circ\circ}$ since $\|a^{kn}|A\| \xrightarrow{n \rightarrow \infty} 0$.

Problem 4

If $\|\cdot|A\|$ is multiplicative, the boundedness of the set $\{a^n \mid n \in \mathbb{N}\}$ is equivalent to the boundedness of $\{\|a|A\|^n \mid n \in \mathbb{N}\}$. And the latter is equivalent to $\|a|A\| \leq 1$. Similarly $\lim_{n \rightarrow \infty} a^n = 0$ is equivalent to $\|a^n|A\| \rightarrow 0$ and by the multiplicativeness this is equivalent to $\lim_{n \rightarrow \infty} \|a|A\|^n = 0$. This is equivalent to $\|a|A\| < 1$.

Problem 5

Take $a \in K$ such that $|a| \in (0, 1)$. Then for a K -Banach algebra A , we have

$a \cdot 1_A$ is a topologically nilpotent element. As any ideal I contains 0, thus if I is open, it contains an open neighbourhood of 0. This means that for sufficiently large n we have $(a \cdot 1_A)^n \in I$. Thus $1_A \in I$ meaning I is not proper.

Problem 6 Let A be a K -Banach algebra. Clearly $1_A \in A^\times$. If $A^\times$ contains an open neighbourhood U of 1, then for any $a \in A^\times$, aU is an open neighbourhood of a contained in $A^\times$. We set,

$$U = \{x \in A \mid \|x|A\| < 1\}.$$

Clearly U is open. Note that for any $a \in A$, $x \mapsto a + x$ is a continuous function with inverse $x \mapsto -a + x$. And as A is a K -Banach algebra, this is a bounded linear operator. Thus $1 + U$ is homeomorphic to U . Consider,

$$(1 - x)(1 + x + x^2 + \cdots + x^{n-1}) = 1 - x^n.$$

As $\|x|A\| < 1$, thus the series,

$$1 + x + x^2 + \cdots$$

is an element of A , we conclude $1 - x \in A^\times$. In other words, $1 + U$ is an open set contained in $A^\times$. Thus $A^\times$ is open.

Problem 7 By Problem 6, we know that in a K -Banach algebra A , $A^\times$ forms an open neighbourhood of 1. If an ideal I in A is dense, I intersects with $A^\times$. This means I contains a unit thus $I = A$.